Roll No. : ....

MID SEMESTER EXAMINATION SEPT. 2025

Name of the Program: **B.Tech**

Name of the Course : **Machine Learning**

Time : 90 Minutes

Semester: **V**

Course Code: **TCS-509**

Maximum Marks : 50

Note:

i. Answer all the questions by choosing any one of the sub questions
ii. Each questions carries 10 marks

Q1	(10 Marks)	CO1
(a)	Define **inductive bias**. Why is inductive bias essential in Machine Learning? Give an example where different biases lead to different hypotheses.	
	OR	
(b)	How a SVM solves a binary classification problem is solved i. Explain the role of the hyperplane and margin in classification. ii. Draw a simple 2D example with support vectors and show how the decision boundary is chosen.	
Q2	(10 Marks)	CO2
(a)	You are given the following dataset with two features (X1, X2) and class labels: Point / X1 / X2 / Class P1 / 1 / 2 / A P2 / 2 / 3 / A P3 / 3 / 3 / B P4 / 6 / 5 / B P5 / 7 / 7 / B A new data point **Q(3,4)** needs to be classified using the **K-Nearest Neighbor (KNN)** algorithm. (a) Compute the **Euclidean distance** from Q to each data point (P1–P5). (b) Classify Q for **k = 3 and k=4**. (c) If **Distance-Weighted KNN** is used (weight = 1/distance), what will be the predicted class?	
	OR	
(b)	Explain **Bias-Variance Tradeoff** with the help of a diagram. How does it relate to the problems of **overfitting and underfitting**?	
Q3	(10 Marks)	CO2
(a)	With the help of a neat diagram, explain the working of the **Version Space**	

The dataset table in Q2(a):

Point	X1	X2	Class
P1	1	2	A
P2	2	3	A
P3	3	3	B
P4	6	5	B
P5	7	7	B

	Algorithm in concept learning. Illustrate your answer with an example data set (e.g., deciding whether to play tennis based on weather conditions).	
	OR	
(b)	The following dataset represents the values of a dependent variable y corresponding to an independent variable x: x — y 1 — 3 2 — 4 3 — 2 4 — 4 5 — 5 1. Explain the concepts of **simple linear regression** and **multiple linear regression** in detail. 2. Using the above dataset, fit a **simple linear regression model** of the form: $y=\beta 0+\beta 1x+\varepsilon$, and compute the **Mean Absolute Error (MAE)** and the **Mean Squared Error (MSE)** of the fitted model.	
Q4	(10 Marks)	CO1
(a)	Discuss various performance measures used by supervised and unsupervised machine learning models with their formula, preferences according to use and importance.	
	OR	
(b)	**What is logistic regression, and what cost function is used to train a logistic regression model? Explain its intuition. How can overfitting in a logistic regression model be handled?**	
Q5	(10 Marks)	CO1
(a)	Discuss different types of cross validation techniques in details.	
	OR	
(b)	Explain the perspectives and issues of Machine Learning in details. Also explain the evolution of machine learning.	CO2